

Supplementary Material: The Effect of Expertise Signaling when Sharing Medical Information in Opinion Dynamics

Anonymous submission

Organization of Supplementary Material

This supplementary material contains additional information for our paper ('The Effect of Expertise Signaling when Sharing Medical Information in Opinion Dynamics'). In this document, we complement the theoretical results with proof, provide details on virtual experiment design, demonstrate the social impact of studied scenarios, and present additional virtual experiments simulating social networks with malicious users and influencers.

Proofs of Theoretical Results

Mathematical Notation

To analytically characterize the multi-agent dynamics, we use the standard definitions and results in Graph Theory and Linear Algebra (see, e.g., Bullo (2019)). For clarity, we first review the most critical definitions and introduce the adopted notation.

Let $A \in \mathbb{R}^{n \times n}$ be a real-value square matrix. Then, $\lambda \in \mathbb{C}$ is a left (right) eigenvalue with $v \in \mathbb{C}^n$ a left (right) corresponding eigenvector iff $Av = \lambda v$ ($w^T A = \lambda w^T$). We denote the spectrum of A (i.e., the set of eigenvalues) by $\text{spec}(A)$ and the spectral radius of A (i.e., maximum norm of eigenvalues) by $\text{spec}(A)$.

We now consider an associated discrete-time linear system $x(k+1) = Ax(k)$, with some initial values $x(0)$ (or x_0). The solution at time k is given by the modal expansion (i.e., $x_k = \sum_i \lambda_i^k v_i(w_i^T x_0)$). If A has non-negative entries and unit row sums (as in a DeGroot model (DeGroot 1974)), we say the associated system is an averaging system. For a convergent averaging system with only one eigenvalue $\lambda_1 = 1$, the final state is determined by the first mode: $\lim_{k \rightarrow \infty} x_k = v_1(w_1^T x_0)$. For a DeGroot model, we highlight the definitions of consensus and stubborn agents below. We also note that when A ensures convergence to consensus, the asymptotic rate of convergence is determined by the largest magnitude of the non-consensus eigenvalues (i.e., $\rho_{\text{ess}}(A) = \max\{|\lambda| \mid \lambda \in \text{spec}(A) \setminus \{1\}\}$), with smaller value indicating faster convergence.

Definition 1 (Consensus). *The state where all opinions are equal, that is, $\lim_{k \rightarrow \infty} x(k) = c \cdot \mathbf{1}$ for some scalar $c \in \mathbb{R}$*

Definition 2 (Stubborn Agents). *An agent i is stubborn if their opinion remains constant (i.e., $x_i(k+1) = x_i(k)$ for*

all $k \geq 0$). Within the DeGroot model, this is modeled by setting the agent's self-influence to one, $a_{ii} = 1$, which implies $a_{ij} = 0$ for all $j \neq i$ due to the row-stochasticity.

We denote the set of all stubborn agents as $\mathcal{S}$ and the set of all non-stubborn agents as $\mathcal{N}$.

A square matrix A of size n has an associated directed graph (i.e., digraph); its vertices are $1, 2, \dots, n$ and edges (i, j) correspond to non-zero entries a_{ij} in the matrix. As usual, paths in digraphs are ordered sequences of nodes such that any pair of consecutive nodes in the sequence is an edge. If a path starts and ends at the same node, we call it a cycle. The in-degree (out-degree) of a node is the number of incoming (outgoing) edges. A node is a source (sink) if its in-degree (out-degree) is zero. We call a node that can be reached from any other node by a directed path a globally reachable node. A graph is strongly connected if there exists a directed path from any node to any other node; this is equivalent to the associated matrix A being irreducible (i.e., $\sum_{k=0}^{n-1} A^k > 0$). A digraph is aperiodic (periodic) if the greatest common divisor of the lengths of all its cycles is (larger than) 1.

Reaching Consensus on a Fixed Network

Our first result considers the scenario where all stubborn users hold one opinion. One such example could be when only experts are stubborn and all agree on one opinion (e.g., 1). Theorem 1 says that if the graph is sufficiently connected (more precisely, contains a following path from every non-stubborn agent to some stubborn agent), then all agents will eventually agree with the opinion of experts.

Theorem 1 (Consensus on a Fixed Network). *Consider a DeGroot system with a non-empty set of stubborn agents $\mathcal{S}$ holding the same opinion c . Assume there exists a directed path from every non-stubborn agent in $\mathcal{N}$ to a stubborn agent in $\mathcal{S}$. Then the system converges to a consensus, with all agents adopting the opinion c :*

$$\lim_{k \rightarrow \infty} x(k) = c \cdot \mathbf{1}$$

The proof of Theorem 1 uses some well-established results. We include them as lemmas below before proceeding to prove the theorem.

Lemma 1 (Consensus for row-stochastic matrices (Bullo 2019) p. 67, Theorem 5.1). *An opinion-dynamic system $x(k+1) = A \cdot x(k)$ converges to a consensus if and only if the weighted digraph associated with the matrix A contains a globally reachable node, and the subgraph of globally reachable nodes is aperiodic.*

Lemma 2 (Convergent Row-substochastic Matrices (Bullo 2019) p.58, Corollary 4.13). *Let A be a row-substochastic matrix. If the digraph associated with A contains directed paths from each node with an out-degree of 1 to a node with an out-degree less than 1, then A is convergent.*

Lemma 3 (Steady State of Affine Systems (Bullo 2019) p.35). *Consider the discrete-time affine system $x(k+1) = Ax(k) + b$. If the matrix A is convergent, then the system has a unique equilibrium point $x^* = (I - A)^{-1}b$, to which all trajectories converge.*

We use the results above to prove Theorem 1.

Proof of Theorem 1. Let the set of all agents be partitioned into non-stubborn agents $\mathcal{N}$ and stubborn agents $\mathcal{S}$ as per Definition 2. By reordering the agents, the influence matrix A can be written in the block upper-triangular form:

$$A = \begin{bmatrix} A_{\mathcal{N}\mathcal{N}} & A_{\mathcal{N}\mathcal{S}} \\ \mathbf{0} & I \end{bmatrix}$$

The dynamics of the system can be decomposed into two. One, for the stubborn agents, whose opinions remain constant

$$x_{\mathcal{S}}(k+1) = Ix_{\mathcal{S}}(k) \implies x_{\mathcal{S}}(k) = x_{\mathcal{S}}(0) = c \cdot \mathbf{1}$$

and another one for the non-stubborn agents,

$$x_{\mathcal{N}}(k+1) = A_{\mathcal{N}\mathcal{N}}x_{\mathcal{N}}(k) + A_{\mathcal{N}\mathcal{S}}x_{\mathcal{S}}(0) \quad (1)$$

The fact that every non-stubborn agent in $\mathcal{N}$ has a path to a stubborn agent, either directly or indirectly through other agents in $\mathcal{N}$, and since the original matrix A is row-stochastic, this means the row sums of the submatrix $A_{\mathcal{N}\mathcal{N}}$ must be less than or equal to 1. That is, $\sum_{j \in \mathcal{N}} (A_{\mathcal{N}\mathcal{N}})_{ij} \leq 1$

Also, the fact that every non-stubborn agent has a directed path to a stubborn one ensures that such a sub-stochastic matrix has a spectral radius strictly less than 1. Therefore, $A_{\mathcal{N}\mathcal{N}}$ is convergent: $\rho(A_{\mathcal{N}\mathcal{N}}) < 1$

Since $A_{\mathcal{N}\mathcal{N}}$ is convergent, we can apply Lemma 3 to the affine system (1).

$$x_{\mathcal{N}}^* = (I - A_{\mathcal{N}\mathcal{N}})^{-1} A_{\mathcal{N}\mathcal{S}} x_{\mathcal{S}}(0) \quad (2)$$

Substituting $x_{\mathcal{S}}(0) = c \cdot \mathbf{1}$, we get:

$$x_{\mathcal{N}}^* = c \cdot (I - A_{\mathcal{N}\mathcal{N}})^{-1} (A_{\mathcal{N}\mathcal{S}} \mathbf{1})$$

Since the original matrix A is row-stochastic,

$$A_{\mathcal{N}\mathcal{S}} \mathbf{1} = (I - A_{\mathcal{N}\mathcal{N}}) \mathbf{1}$$

Substituting back in (2),

$$x_{\mathcal{N}}^* = c \cdot (I - A_{\mathcal{N}\mathcal{N}})^{-1} (I - A_{\mathcal{N}\mathcal{N}}) \mathbf{1} = c \cdot \mathbf{1}$$

Since the dynamic of both stubborn and non-stubborn agents converge to the same final value, the entire system reaches a consensus state.

To fully comply with the Lemma 1, aperiodicity follows from stubbornness, since stubbornness is a self loop or a cycle of size 1. Meaning that its greatest common divisor can only be 1, satisfying the definition for aperiodicity. $\square$

Impact of the Weights on the Convergence Speeds

Next, we consider what happens to the speed of convergence as the trust in stubborn users increases. This would correspond to an intervention that makes experts more visible or trustworthy. The online experiment suggests that one such intervention could be to encourage them to signal their expertise. Based on Theorem 2, such interventions would increase the speed needed for reaching a consensus.

Theorem 2 (Impact of Weights on the Convergence Speeds). *Consider a DeGroot system, $x(t+1) = Ax(t)$, with a non-empty set of stubborn agents $\mathcal{S}$ holding the same opinion c . Assume there exists a directed path from every non-stubborn agent in $\mathcal{N}$ to a stubborn agent in $\mathcal{S}$. Let a second influence matrix A' be formed by increasing the influence of $\mathcal{S}$ on $\mathcal{N}$. The convergence of the non-stubborn agents under A' is at least as fast as under A :*

$$\rho(A'_{\mathcal{N},\mathcal{N}}) \leq \rho(A_{\mathcal{N},\mathcal{N}})$$

Furthermore, when the influence is strictly increased (i.e., $A' \neq A$), the convergence is strictly faster.

The proof of the theorem uses the following result from the literature:

Lemma 4 (Monotonicity of Spectral Radius (Bullo 2019) p.56, Corollary 4.10). *Let A and A' be two non-negative matrices. If $A \leq A'$, then $\rho(A) \leq \rho(A')$. If additionally $A \neq A'$ and A' is irreducible, then the inequality is strict: $\rho(A) < \rho(A')$.*

Proof of Theorem 2. We use the same block upper-triangular notation for the influence matrix A (obtained by possibly reordering agents) as in the proof of Theorem 1:

$$A = \begin{bmatrix} A_{\mathcal{N}\mathcal{N}} & A_{\mathcal{N}\mathcal{S}} \\ \mathbf{0} & I \end{bmatrix}$$

The dynamics of the system can be decomposed into two. One, for the stubborn agents, whose opinions remain constant

$$x_{\mathcal{S}}(k+1) = Ix_{\mathcal{S}}(k) \implies x_{\mathcal{S}}(k) = x_{\mathcal{S}}(0) = c \cdot \mathbf{1}$$

and another one for the non-stubborn agents,

$$x_{\mathcal{N}}(k+1) = A_{\mathcal{N}\mathcal{N}}x_{\mathcal{N}}(k) + A_{\mathcal{N}\mathcal{S}}x_{\mathcal{S}}(0)$$

By construction, the influence of the stubborn set $\mathcal{S}$ on the non-stubborn set $\mathcal{N}$ is increased in A' . An increase in stubborn influence means $\sum_{k \in \mathcal{S}} (A'_{\mathcal{N}\mathcal{S}})_{ik} \geq \sum_{k \in \mathcal{S}} (A_{\mathcal{N}\mathcal{S}})_{ik}$.

For any non-stubborn agent $i \in \mathcal{N}$, the row-stochastic property of A and A' requires:

$$\begin{aligned} \sum_{j \in \mathcal{N}} (A_{\mathcal{N}\mathcal{N}})_{ij} + \sum_{k \in \mathcal{S}} (A_{\mathcal{N}\mathcal{S}})_{ik} &= 1 \\ \sum_{j \in \mathcal{N}} (A'_{\mathcal{N}\mathcal{N}})_{ij} + \sum_{k \in \mathcal{S}} (A'_{\mathcal{N}\mathcal{S}})_{ik} &= 1 \end{aligned}$$

This requires a decrease in the non-stubborn row sums: $\sum_{j \in \mathcal{N}} (A'_{\mathcal{N}\mathcal{N}})_{ij} \leq \sum_{j \in \mathcal{N}} (A_{\mathcal{N}\mathcal{N}})_{ij}$.

This implies element-wise $A'_{\mathcal{N}\mathcal{N}} \leq A_{\mathcal{N}\mathcal{N}}$.

By applying Lemma 4 to this inequality, we obtain:

$$\rho(A'_{\mathcal{N}\mathcal{N}}) \leq \rho(A_{\mathcal{N}\mathcal{N}})$$

That is, increasing the influence of stubborn agents leads to a lower convergence factor.

Additionally, if $A'_{\mathcal{N}\mathcal{N}} \neq A_{\mathcal{N}\mathcal{N}}$ and the associated graph of the submatrix $A_{\mathcal{N}\mathcal{N}}$ is strongly connected, which, as mentioned before, is equivalent to being irreducible. Lemma 4 also guarantees that the convergence factor is strictly smaller, implying strictly faster convergence. $\square$

Altogether, Theorems 1 and 2, show that when the graph representing the social network is sufficiently connected (with a following path from every common user to an expert) all users will eventually agree with the experts. Moreover, the speed of reaching that point can be increased by raising the levels of trust of at least one common user towards an expert.

Corollary 1. *Consider a DeGroot system, $x(t+1) = Ax(t)$, with a non-empty set of stubborn agents $\mathcal{S}$ holding the same opinion c . Assume there exists a directed path from every non-stubborn agent in $\mathcal{N}$ to a stubborn agent in $\mathcal{S}$. Let a second influence matrix A' be formed by (strictly) increasing the influence of $\mathcal{S}$ on $\mathcal{N}$. If both systems have the same initial condition, their vectors of opinion converge to the same consensus c , with the only difference that the system defined by A' converges faster.*

Data-driven opinion dynamics model

Modeling different types of agents

As mentioned in the main paper, we extend the DeGroot model by distinguishing different types of actors on social media. In the following simulation experiments, we defined four different types of agents, namely *common users*, *doctors*, *malicious users*, and *influencers*. As presented in Table 1, these agents differ based on their stubbornness, title, and initial opinion. These are all important dimensions for the opinion dynamics: our experiment showed that expertise signaling can affect trust¹, while other prior work underlines the importance of malicious users (including bots) and influencers (high in-degree centrality nodes) in shaping the online discourse (Ferrara et al. 2016; Thompson 2023).

As such, *common users* are the general population on the social media platform who update their opinions at each timestep. *Doctors* are the experts whom we assume to hold one opinion universally accepted as true across experts. *Malicious users* refer to users who intentionally disseminate false or misleading information to deceive public perception and cause harm, such as trolls, bots, or impersonators. *Influencers* are users who possess substantial followings, thus we define them as the top 2% of users who own most of the followers.

¹As there was no significant difference between 'Dr.' and 'MD' revealed from the online experiment, we only study one of them ('Dr.') with simulation. One can presume the other ('MD') has a similar effect for the interventions tested in the simulation part.

Table 1: Overview of the different types of agents in our model. For each type, we specify: if the agent is *stubborn* (i.e., keeps their initial opinion fixed), the professional *title* they could include in their display, and the range of values for their *initial opinion*.

Agent type	Stubborn	Title	Initial opinion
Common user	False	None	[0, 1]
Expert	True	'Dr.' / None	1
Malicious user	True	'Dr.' / None	0
Influencer	True	'Dr.' / None	1

Informing Parameters for Simulations

The model introduced above is based on a couple of parameters, which we overview in Table 2. For the simulations, we ground the parameter choices in our online experiment data and prior work. We provide an overview below and leave the details for Section Simulation in the Appendix.

Table 2: Tabular description of model parameters (Left), the values taken in the simulations (Center), and the virtual experiments for which these values are varied (Right). We use bold (red) for default parameter values: i.e., we use this value when the parameter in question is not varied. For each combination of parameters, we run 100 simulations.

Varied Parameters	Values Taken	Vary in
% of experts with title	0, 20, ..., 100	E1, E4
Number malicious users	0 , 100, ..., 400	E2
% titled malicious users	0 , 25, ..., 100	E2
% titled influencers	0 , 25, ..., 100	E3
Network	ER (edge pr. 0.01), SF ($\alpha = 2.5$) PolBlog	E1, E2
Fixed Parameters	Value	
Total number of agents	1500 (Adamic and Glance 2005)	
Number of experts	100	
Opinion of experts	1	
Opinion malicious users	0	
Number of influencers	30 (top-degree 2%)	

Initial opinions. We base the initial opinion of common users on the participants' initial trust in the medical statements (i.e., their statement-agreement during pre-experiment). To do so, we first map the 7-Likert scale to [0, 1]. This gives a normal opinion distribution with mean 0.54 and standard deviation 0.29. Thus, during simulations, we sample the initial opinions of common users from a normal distribution $\sim \mathcal{N}(0.54, 0.29)$ (truncated above 0 and below 1).

Weight Matrix. We ground the weight matrix in the observed trust of participants in posts sent by different accounts (with or without a title in the display name). The experiment corresponds to a simplified case of opinion dynamics: one participant, i , updates their opinion after seeing a message from a sender, j , based on their trust in themselves, w_{ii}^* , and their trust in the sender, w_{ij}^* :

$$\begin{cases} x_i(1) = w_{ii}^* x_i(0) + w_{ij}^* x_j(0) \\ w_{ii}^* + w_{ij}^* = 1 \\ x_j(0) = 1 \end{cases} \quad (3)$$

where $x_i(1)$, $x_i(0)$, and $x_j(0)$ are respectively the initial opinion of participant from the pre-experiment survey, the changed opinion of participant from the post-experiment survey, and the message sender’s opinion. These values are known for each participant i , allowing us to solve System (3) and find w_{ij}^* . By taking in turn averages when the sending account has a title in the display name or not, we obtain an average trust in titled users of $w_{i,Dr}^* = 0.183$ and common users of $w_{i,CU}^* = 0.003$. For simulations, we follow the approach of Zhou et al. (2020) and preserve row stochasticity by normalizing the trust based on each agent’s connectivity.

Networks. We use two standard models for generating synthetic social networks, namely Erdős–Rényi (ER) and Barabási–Albert. The ER model creates social networks in which each node has roughly equal influence, corresponding to communities of people with similar interests or occupations (Newman, Strogatz, and Watts 2001). Each user on the ER network shares uniform information exposure. The Barabási–Albert model creates scale-free (SF) networks. These reflect the typical connectivity of real-world social media platforms, where a few high-degree nodes (influencers) dominate visibility and diffusion (Mislove et al. 2007; Eikmeier and Gleich 2017). However, Broido and Clauset study also suggests that real-world social networks are mostly weak scale free (Broido and Clauset 2019). To avoid deviating from reality, we also used a political blog network (PolBlog) in some experiments (Adamic and Glance 2005). This network is widely used in research on information diffusion (Peralta et al. 2021; Peralta, Kertész, and Iñiguez 2022; Zhang et al. 2025), especially in studies based on the DeGroot model (Zhang et al. 2025).

Evaluating Social Impact

Evaluation metrics. In order to evaluate the social impact of signaling expertise on social media platforms, we adopt the standard metrics in the opinion dynamics literature. First, we investigate if a *consensus* (i.e., a state where opinions do not change anymore and all agents hold the same opinion) is reached. If this is the case, we also look at the *time needed to reach consensus* and the *consensus value*. These metrics are crucial as they capture whether communities of users will eventually agree, what they would agree on, and how long it takes for this agreement to arise. In the context of medical information, consensus among scientists and public consensus around expert-backed evidence is socially beneficial, as it promotes informed decision-making and reduces the influence of misinformation (van der Linden, Clarke, and Maibach 2015; Maibach and Van Der Linden 2016).

However, not all discrete-time linear systems converge to a consensus (Degroot 1974; Berger 1981; Dong et al. 2017) and many real-life platforms remain polarized (Schmidt et al. 2018; Jiang, Ren, and Ferrara 2021). In such cases, we consider the *average user opinion* over time. Refer to

Das, Gollapudi, and Munagala (2014)’s work, we also investigate the level of polarization by looking at the proportion of agents whose final opinions are close to the two extreme values (here operationalized as either above 0.8 or below 0.2, representing those who find the statement likely or respectively unlikely to be true). Altogether, these measures enable us to assess the societal effect of titles in terms of collective opinion, individual belief strength, and temporal dynamics.

Interventions. We aim to investigate how platform policy and design choices concerning the signaling of expertise can affect the opinion dynamics. As shown in Table 2, these different platform choices correspond to different parameter values in our model. First, platforms could encourage more experts to signal their expertise in their display name (Majid et al. 2019); this corresponds in our model to varying the percentage of doctors that include their title. Previous works have proved the effect of malicious user removal (Lee and Kim 2014) and identity verification (Pang, Pavlou et al. 2023) in mitigating misinformation; in our model, we vary the number of malicious users and the percentage of titled ones. Social media influencers can help promote scientific information (Galetti and Costa-Pereira 2017); we compare two cases by varying the number of influencers perceived as credible as doctors: influencers post information themselves or repost from experts. Key opinion leaders (KOL) among the doctors are usually with higher titles and can be identified by recommender systems (RS) (Guo et al. 2016). Moreover, expert identification and recommendation may have positive impact on users’ behaviors (Herm-Stapelberg and Rothlauf 2020; Sha et al. 2012); we further improve the visibility of one expert with the RS by feeding their opinion to all users.

Setting and assumptions. While both our model and code for the simulation are general and could capture a variety of settings, we need to make some assumptions in order to maintain the feasibility of the analysis. More precisely, we assume experts unanimously agree on one opinion (without loss of generality, the one with value 1). As usual, we assume malicious users try to promote the opposite opinion (Egele et al. 2013). In one of the experiments, we assume all influencers assist in promoting the verified information. Additionally, when simulating the effect of RSs, we assume all doctors have already signaled titles. We also assume that there are more bots than doctors on social media (Ng and Carley 2025).

Virtual Experiments

While the analytical results specify the direction of change resulting from interventions, the magnitude of these changes remains unknown. Therefore, we use simulations to find the impact of interventions. Beyond encouraging doctors to signal title in their username in Experiment 1 (E1) shown in the main paper, we conducted 3 additional experiments to explore the interventions under more complicated scenarios. The paragraphs below introduce the experimental setups for all four of our simulations. Table 2 shows an overview of the different parameter combinations.

Experiment 1 (E1) verifies the effectiveness of encouraging more experts to signal their expertise. As such, we vary the percentage of ‘Dr.’ signaling their expertise (i.e., 0, 20, 40, 60, 80, 100). We evaluate the convergence speed and average opinion in both ER and SF networks.

Experiment 2 (E2) captures situations where doctors promote verified medical information and malicious agents (e.g., bots) promote misinformation while possibly posing as experts. We consider two interventions targeting these malicious agents: decreasing their number (e.g., via effective bot detection or moderation) and verifying expertise (leading to lower numbers of malicious agents posing as experts). To capture this setting, we vary the number of malicious agents (between 0 and 400) and the percentage of these agents using a title in their display name. Since consensus does not emerge, we consider the average opinion and percentages of users with high trust in one of the extreme opinions (i.e., above 0.8 or below 0.2).

Experiment 3 (E3) investigates the role of influencer-related interventions. Prior work shows that highly connected nodes (i.e., influencers) have a big impact on the opinion dynamics (Dong et al. 2017; Helfmann et al. 2023). We also carry a pre-experiment which shows the same is true in our model: In SF networks with few nodes of high degree (i.e., influencers), the initial opinion of influencers has a higher impact on the final average opinion than that of experts, who largely have lower connectivity (details in Section Simulation of the Appendix). We then investigate what happens as we encourage influencers to promote verified medical information, either as themselves or by reposting the posts of experts. For this, we take the top 30 most followed agents as the influencers and assign them the opinion of experts, i.e. 1. We assume that when reposting messages from experts, influencers are perceived as credible as titled users. Therefore, we test the intervention by varying the number of titled influencers (from 0 to all 30).

Experiment 4 (E4) investigates the effect of promoting the opinion of experts by increasing the visibility of their posts algorithmically, i.e., intervening on the recommender system (RS) rather than encouraging influencers to post the opinions of experts as in E3. Since opinion dynamics models only capture one piece of information, we cannot implement state-of-the-art RSs such as collaborative filtering. Instead, similarly to Sprenger et al. (2024), we randomly select one doctor agent corresponding to a verified expert whose opinion the platform would want to promote from a normative perspective. Next, we add a following relationship from all the common users to the selected doctor at the beginning of the simulation. As explained by Sprenger et al. (2024), this selected node acts as an RS-node, capturing the opinion promoted by the RS. We compare the cases where this selected expert have signalled title or not. Furthermore, we also compare both cases with influencer repost from experts in E3.

Simulation Results

E1: Convergence time and average opinion increase as more experts signal their expertise

Fig. 1 (a) shows that on ER networks the final opinion reach a consensus at 1. Moreover, the time for reaching consensus (respectively 2022.76, 762.74, 582.29, 522.59, 496.55, 470.64 steps) greatly reduces from the setting with no doctors signaling their expertise to the one when 20% do so (from 2023 to 763 steps). Further increasing this percentage produces diminishing gains in convergence speed. In Fig.1 (b), we plot only 3 conditions for E1 on the SF network: 0, 50, 100 doctors have ‘Dr.’ displayed in usernames. Although the average opinions increase when more doctors signal titles, the differences are not significant with overlapping confidence intervals. By investigating the simulation runs, we believe this is caused by the randomness in the position of experts within the network. More precisely, the gains would be large if the ‘Dr.’ signaling expertise has a high degree and negligible if they are a periphery node (analyses in Section Simulation-Pretest in the Appendix). We plot the same conditions for the PolBlog network in Fig.1 (c). Although the opinions do not converge to 1 as on the ER network, the randomness is less than the SF network and signaling title tends to be effective when fewer doctors are doing so.

Policy implications. The results of E1 suggest that encouraging doctors to signal expertise is an efficient intervention. Throughout scenarios, it can bring the average opinion of users closer to that of experts. In networks with high homogeneity and uniformly random connections, it can greatly reduce the time people need to be fully convinced of the experts’ opinion. Such an intervention is particularly effective when only a few experts already signal their expertise.

E2: Verifying expertise signaling has a higher impact than high-accuracy bot detection

We now look at common users’ opinion while gradually removing malicious users or their titles (e.g., by verifying claims of being an expert) on the ER network with 100 titled experts fixed. The results on the SF network are still random, like in the E1 (see in Appendix, Simulation-Results). Fig.2 shows that both interventions bring the collective opinion closer to 1. For example, with 400 titled malicious users, the average opinion is only 0.2; removing their titles raises this opinion to 0.83, removing 75% of bots increases it to 0.5, while removing all bots eventually brings the opinion to 1.

When removing bots, it also matters which ones are removed. More precisely, our results show that removing untitled bots has little to no effect: e.g., the final average opinion is the same when there are (i) only 200 titled bots and (ii) 200 titled and 200 untitled bots. For E2, we also plot the fraction of common users with opinions close to the extreme values and offer analyses in the Appendix-Simulation.

Policy implications. The above results suggest that both removing malicious users and verifying expertise are effective interventions to mitigate the impact of malicious users. Additionally, when there is evidence indicating that some malicious users pose as doctors, it is more important to identify such users and remove their titles first. As such, when

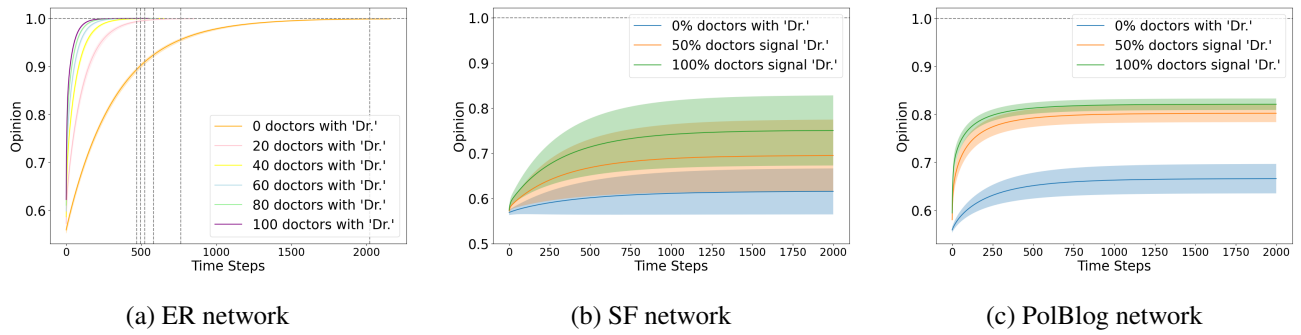


Figure 1: E1 - The average opinion over time for different networks and different shares of experts signaling expertise. Lines show the average over 100 simulations. The shaded areas represent the standard deviation in average opinion across simulations.

resources are scarce, it might be more efficient to prioritise implementing a verification of expertise.

E3: Influencers reposting content from verified experts is more efficient than making original posts

We check the average opinion dynamics when encouraging influencers to repost from doctors instead of posting verified medical information themselves. In Fig. 3, we see that for both interventions the final opinion is close to 0.9 (though, for influencers making original posts, this value is slightly lower). The difference lies in the speed of reaching this final opinion. Especially at the beginning of the simulation, the opinion change is very fast when influencers repost from experts, and more gradual when they make original posts. The same is true when considering the number of users with individual opinions close to the verified one. Around 1200 out of 1400 common users eventually reach opinion 1. When influencers repost from experts, most of them change their opinion within the first 40 steps, while when they send messages as themselves, this process takes around 500 steps.

Policy implications. The results in E3 suggest that influencers reposting the posts of verified experts could be more efficient than directly posting the same message themselves. Such improvement is especially meaningful on contemporary social media platforms with fast-paced information transfer, as rapid consensus-building can be critical for public response.

E4: Improving experts' visibility leads to a consensus to their opinion

Our first theoretical result suggests that ensuring a path from common users to experts guarantees consensus. The results of E4 confirm this. As shown in Fig. 4, the opinions of common users converge to 1 when improving the visibility of doctors. We also gain additional insight from the theoretical work. If the doctor signals expertise with 'Dr.', the common users change their opinions very quickly in 35 steps. If the doctor does not include a title, it takes a longer time (1917 steps) for convergence. Compared with the previous influencer intervention in E3, promoting one doctor's opinion through the recommender system leads to consensus to the doctor's opinion, while promoting through influencers may still leave a small group of people skeptical.

Policy implications. These results suggest that recommending a doctor's opinion may be a more efficient way than encouraging influencers to repost the doctor's opinion. Encouraging doctors to signal their titles in their display name is important to greatly speed up this process.

Conclusion

We investigate how signaling expertise (e.g., including professional titles) shapes trust in medical information and influences long-term opinion dynamics on social media. Through an online experiment, we find that including professional titles (e.g., Dr., MD) in the display name of the source slightly yet significantly increases the perceived credibility of their posts. We use this data to extend the DeGroot model with expertise signaling and base its parameters on the experiment's data. A theoretical analysis of the resulting discrete-time system specifies sufficient conditions for consensus to emerge and demonstrates that encouraging expertise signaling increases the speed of reaching it. Through simulations, we find that on a random network, even a small group of titled users can have a large influence on the aggregate opinion. On a scale-free network, however, simply encouraging experts to signal titles or verifying identity (e.g., removing malicious users or their 'Dr.' titles) may be less effective than interventions through influencers (e.g., encouraging them to promote verified medical information). Results also suggest that an even more effective intervention could be to algorithmically promote the content of experts (e.g., through recommender systems).

Nonetheless, our work remains an initial step and has several limitations that could be tackled by future research. For example, our empirical experiment is exploratory. Future work can use it to conduct a power analysis and build a confirmatory experiment; such work can also verify if our findings apply to more diverse samples of participants. Like any agent-based model, our model makes several simplifying assumptions on user behavior (e.g., binary stubbornness, equal level of trust in non-expert sources, static trust networks). Additionally, we did not consider the time and cost of implementing interventions (e.g., it can be time-consuming for experts to actively participate on social media (Galetti and Costa-Pereira 2017)).

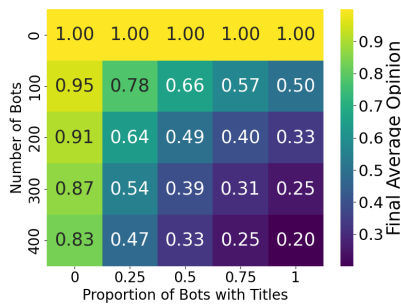


Figure 2: E2 - Heatmap of common users' average opinion. The vertical axis is the number of bots in the network, and the horizontal axis is the percentage of bots with a title.

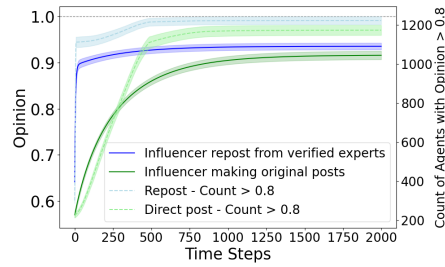


Figure 3: E3 - The average opinion and counts of common users with opinion > 0.8 . The lines represent the average results for 100 simulations, and the shaded areas are built on standard deviations.

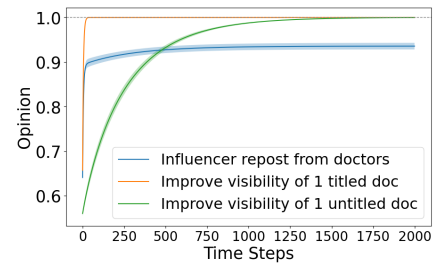


Figure 4: E4 - The dynamics of average opinion when: Influencers repost from doctors; Recommend 1 titled doctor's opinion; Recommend 1 untitled doctor's opinion

Altogether, our work is a first step towards understanding how expertise signaling and related interventions affect medical opinions. At a methodological level, it combines online experiments, theoretical analysis, and agent-based modeling, demonstrating the benefits of a holistic approach. Having a dedicated online experiment allowed us to vary one variable at a time, thus isolating its effect. This represents an alternative to calibrating model parameters based on digital trace data, which is more contextual and has many hard-to-disentangle effects. It also shows how isolating directions of change via dedicated experiments supports a more insightful theoretical analysis, connected to real-life changes and interventions. Thus, more generally, our work demonstrates the potential of experiment-driven modeling for supporting realistic social simulations and an initial comparison over potential interventions.

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Appendix - Simulation

This section includes the pretests we carried out before the virtual experiments, motivation for the parameters chosen for the virtual experiments, and extra results for virtual experiment E2.

Pretest - Opinion Dynamics without Title

With our agent-based model (described in Section Data-driven opinion dynamics model in the main text), we first run three preliminary tests that exclude the effect of title (i.e., all agents are non-title) on both the ER and SF networks. The three preliminaries respectively tested the following scenarios: common users adapt their opinion based on each other (PE1), doctors promoting verified information to the common users (PE2), doctors promoting verified information against malicious bots promoting misinformation (PE3). In PE1, we included 1500 *common users*. In PE2, we included 100 *Doctor* and 1400 *Common User*. In PE3, we included 100 *Doctor*, 1300 *Common User* and 100 *Malicious bot*. For all preliminaries, we ran 2000 time steps with a total number of 1500 agents.

(*Results*) We plot the opinion dynamics of one simulation run for each test. As shown in Fig. 5 and Fig. 6, the grey lines represent the one agent’s opinion, and the black line represents the average opinion change over time. Fig. 5 are the results of three tests on the ER network. When there are no stubborn users or only stubborn users with a unified opinion, the final opinion respectively converges to the average or the stubborn agents’ opinion. When stubborn users hold different opinions, the final opinion clusters around the average opinion but does not converge. On the SF network, as shown in Fig. 6, the final opinion does not tend to converge. The trends of average opinions are more random and suggest that the presence and opinion of stubborn users have little impact on the dynamics.

(*Analysis*) We also investigate why stubborn users have such little impact in SF networks. To do so, we run a simulation on a small part of the SF network and monitor the local opinion dynamics. Fig. 7 shows an example found this way. There, I is a high in-degree node (Influencer with a large number of followers), D represents an expert node, who is stubborn with opinion 1, $C1$, $C2$ and $C3$ represent three random common users. $C1$ is directly influenced by both I and D , $C2$ is indirectly influenced via $C1$, and $C3$ is only affected by I . Together, these four types of nodes broadly capture the typical roles and connection patterns present in SF networks. Assuming uniform edge weights (i.e., ignoring the effect of title), $C1$ ’s opinion gradually moves toward the average of I and D , without fully reaching the doctor’s position. $C2$ ’s opinion eventually converges to $C1$. $C3$ ’s opinion converged to I ’s opinion. This dynamic helps explain why, in SF networks, the collective opinion remains closer to influencers rather than experts. In this example, unless the influencer directly follows at least one expert, experts’ opinions only have a weak influence. Connecting to our analytical results and proofs, the existence of a following path from all common users to at least one expert would guarantee convergence. This analysis revealed that SF networks rarely fit

this criterion. As a result, by choosing experts at random from the network has a high chance of them not being influencers, thus having relatively low levels of connectivity and impact on opinions.

Virtual Experiment Design

Here, we introduce the parameters and simulation details of the four virtual experiments.

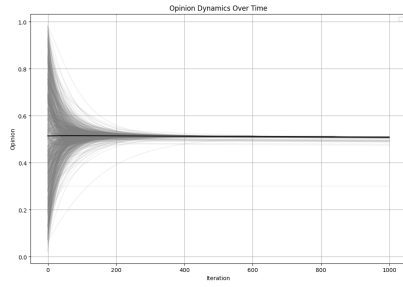
Table 3 is an expanded list for the parameters we introduced in the main paper. For the networks we use, the parameters of ER and SF networks are selected to align with parameters of the real political blog network (Adamic and Glance 2005). We further define the top 2% users with most followers as the influencers on the network according to Helfmann et al. (2023), whose work adopted 250 common users with 4 influencers. For the proportion of medical experts on social media, prior studies have noted that their numbers are relatively low (Ke, Ahn, and Sugimoto 2017). Yet it is hard to make a concrete estimation, as it is hard to verify all the experts on social media. To allow sufficient flexibility in varying the number of experts signaling their titles, we set the total number of experts at a relatively high level (100 out of 1500 agents). According to Glenski, Weninger, and Volkova (2018)’s work, misinformation sources have a higher number of users who retweet each post compared to trusted sources. Therefore, we define the maximum number of malicious users to be larger than the number of experts (400 vs. 100 agents), and vary it in intervals of 100. As we already illustrated in the main paper, Section Evaluating Social Impact, Subsection Setting and assumptions, we assume experts all hold opinion = 1 and malicious users all hold opinion = 0.

Moreover, we also define the end conditions for opinion dynamics. For computational reasons, we stop the simulation run when one of the following conditions is met: 1) opinions reach a consensus (every agent’s opinion deviates from the average opinion by no more than 0.001), 2) opinions reach equilibrium (for all agents, the change of opinion from the previous step is less than $1 \cdot 10^{-7}$), or 3) reach the maximum iteration timestep (2000, chosen based on the pretest). For every condition in the four experiments, we run 100 simulations and calculate the means and standard deviations to avoid reporting outliers or non-representative results. The random seeds used for each run are the index of the simulation (i.e., random seed = 100 for the 100th simulation).

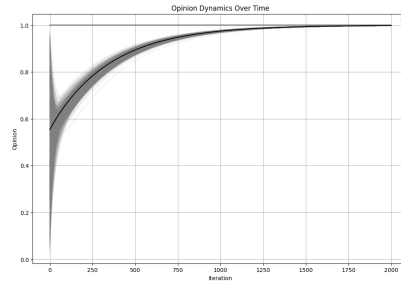
Results

Additional results in E2 In the main paper, we present the results and analysis of E2 run on the ER network. In this part, we offer additional analyses on the percentage of users whose opinion is close to extreme on the ER network, and all the results on the SF network.

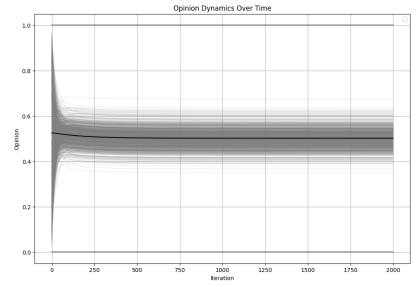
From the average final opinion in E2 on the ER network, we discover that it also matters which ones are removed when removing malicious users. The same applies when considering the fraction of common users with opinions close to the extreme values: Fig. 8 shows that 6% of users lean towards the opinion of experts in both scenarios



(a) PE1 - No stubborn agent

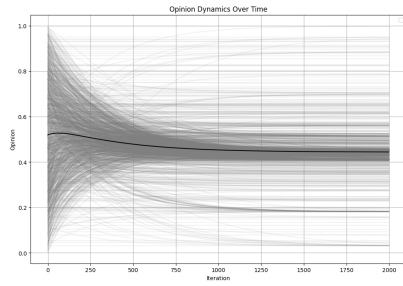


(b) PE2 - 10% stubborn agents with uniform opinion = 1

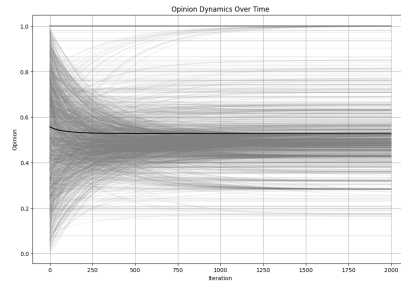


(c) PE3 - 10% stubborn agents with opinion = 1 and 10% with opinion = 0

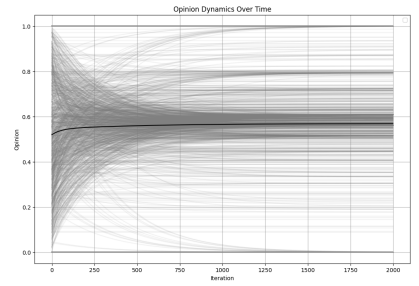
Figure 5: Opinion dynamics of all agents on the ER network. The horizontal axis shows time and the vertical axis shows opinions.



(a) PE1



(b) PE2



(c) PE3

Figure 6: Opinion dynamics of all agents on the SF network.

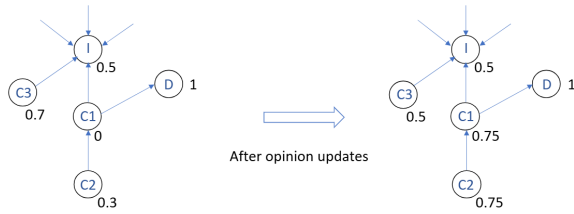
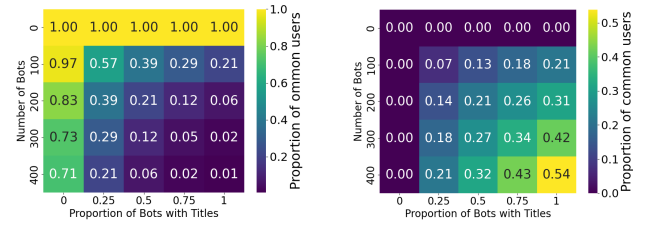


Figure 7: An example of local opinion dynamics on SF network, where I is an influencer agent, D is an expert agent, $C1$, $C2$, and $C3$ are common user agents.

(i) and (ii), while 32% and respectively 31% lean towards the opinion of bots. In Fig. 8, we can see that similar trends exist on the individual level. For example, in the scenario with 100 bots and 50 of them are titled, 39% common users tend to trust the doctors' opinion and 13% tend to trust bots' opinion. While in the scenario with 200 bots and 50 of them titled, the same 39% of common users tend to trust the doctors' opinion and 14% tend to trust bots' opinion.

Additionally, the first columns of Figures 8 show that when no malicious users pose as doctors and all the doctors signal title, no common users tend to completely trust the bots, regardless of how many malicious users there are



(a) proportion of common uses whose final opinion is above 0.8

(b) proportion of common uses whose final opinion is below 0.2

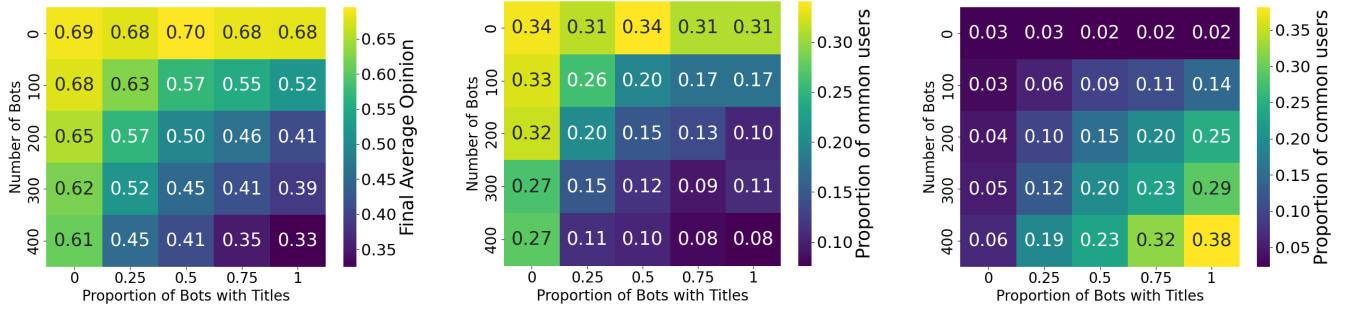
Figure 8: E2 - the proportion of common users who believe the extreme opinions on the ER network (average opinion provided in the main paper). The results only calculate the opinion of common users, with experts and malicious users ruled out.

(low-level opinion distribution in Appendix). Yet removing malicious users will increase the average opinion by convincing more people to trust the doctor's opinion (from 71% to 100%).

We also look into the average and individual final opinions on the SF network in E2. The average results of 100

Table 3: This table is an extension of Table 3 in the main paper. Besides the varied and fixed parameters, we also introduce how these parameters are selected.

Varied parameters	Value	Selection criterion
% of experts with title	0, 20, ..., 100	(Glenski, Weninger, and Volkova 2018)
number of malicious users	0 , 100, ..., 400	
% titled malicious users	0 , 25, ..., 100	
% titled influencers	0 , 25, ..., 100	
Network	ER (edge pr. 0.01), SF ($\alpha = 2.5$), PolBlog	aligned with PolBlog aligned with PolBlog (Adamic and Glance 2005)
Fixed parameters	Value	Selection criterion
Overall time steps	max iteration (2000), or consensus (0.001), or equilibrium ($1 \cdot 10^{-7}$)	based on estimated time steps
Simulation runs in 1 experiment	100 simulations	(Wang and Wu 2024)
Random seeds	1-100	index of simulation runs
Total number of agents	1500	(Adamic and Glance 2005)
Number of experts	100	
Opinion of experts	1	
Opinion of malicious users	0	
number of influencer	30 (2%)	(Helfmann et al. 2023)



(a) the final average opinion of all common users.

(b) proportion of common users whose final opinion is above 0.8.

(c) proportion of common users whose final opinion is below 0.2.

Figure 9: E2 - the proportion of common users who believe the extreme opinions on the SF network. The results only calculate the opinion of common users, with experts and malicious users ruled out.

simulations are offered in 9. In general, the average opinions and percentage of common users with opinion above 0.8 go up while removing bots or their titles. However, between two neighboring conditions, the results can be random sometimes (e.g., 300 malicious users 75% with titles v.s. 300 malicious users 100% with titles). When we look at the percentage of common users with opinions below 0.2, the trends are more consistent while removing malicious users and their titles. Considering we run 100 simulations to avoid randomness in single results, the above results may suggest: even though the effect is less prominent compared with it on the ER network, both removing malicious users and removing their titles can still reduce the influence of the misinformation. Such influence is more evident in reducing the proportion of common users who trust the malicious users' opinions.

Appendix - Discussion

Different types of expertise signaling. An interesting finding in our empirical study is that titles have a significant impact on the perceived credibility, while profiles signaling expertise do not. One possible explanation is that titles within display names are more prominent. They quickly convey expertise, and viewers do not check them across bios. Please note that our experiment only tested the profile variable in conditions with titled display names. This could have made the expertise of bios redundant, as part of that information was already conveyed in display names. As such, the results indicate that once a title is included in the display name, the level of expertise-related information in the bios does not further influence the credibility of posts. Future experiments could investigate the individual effects of information within bios as well as the reasons why display names have a higher influence on credibility.

Practical considerations for implementing interventions.

Results of both Experiment 1 and Experiment 2 suggest that even partial compliance with platform guidelines is enough to affect the opinion dynamics: even a limited number of doctors or malicious users signaling expertise can prominently shift the average opinion. Our results show that the gains from interventions depend on the current proportion of titled users within the target group (e.g., experts, malicious users). Beyond this, the results in E2 also offer a potential response to the concern about deleting accounts (Atreja, Hemphill, and Resnick 2023). Atreja, Hemphill, and Resnick (2023) consider account removal as a high-severity intervention, which may lack legitimacy, cause silencing of dissent and lead to migration to other platforms. In practice, promoting information by encouraging scientists to post on social media can be difficult (e.g., as communicating with the public can cost time), therefore having influencers promote the information may be more practical (Galetti and Costa-Pereira 2017). Experiment 3 confirms this intervention to be effective in our setting too, showing that influencers delivering information from verified experts may speed up the opinion dynamics.

An alternative is promoting the opinion of experts algorithmically, e.g., via recommender systems (RS). This can reduce the time costs needed by expert to form their networks. The results of Experiment 4 show that this intervention would improve the opinion dynamics, especially when combined with an increased signaling of expertise. To implement such an intervention, platforms could rely on prior work that identifies key opinion leaders among doctors based on expertise (Guo et al. 2016). In terms of verifying accounts, platforms such as YouTube have already started to provide labels (e.g., ‘From a doctor licensed in the US’) underneath the display name². Altogether, our results suggest that starting and extending such efforts to emphasize and verify experts’ professional credentials, as well as increasing their visibility on topics where they have more expertise, could support better opinion dynamics.

²<https://support.google.com/youtube/answer/9795167>